

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): LX720, LW720, LH720

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.01.0000-b010	2.1.0

Version History

Module Version	Date	Notes
1_0_0_0	2/7/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LX720')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LX720')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 8000, Model='LX720')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command 3DMode	Value 'Auto' 'Frame Packing' 'Inverter Disabled'	Value 'Top and Bottom' 'Side by Side' 'Inverter Enabled'	Value 'Frame Sequential' 'Sync Off'
# 3DMode example InterfaceName.Set('3DMode', 'Auto')			
Command AspectRatio	Value '4:3' 'Auto'	Value '16:9' 'Real'	Value '16:10'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage ¹	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'PC 1 / YPbPr 1' 'HDMI 2 / MHL'	Value 'PC 2 / YPbPr 2' 'Video'	Value 'HDMI 1' 'S-Video'
# Input example InterfaceName.Set('Input', 'PC 1 / YPbPr 1')			
Command LampMode	Value 'Normal' 'Custom'	Value 'Economic'	Value 'Dimming'
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command	Value	Value	Value

MenuNavigation	'Up' 'Right' 'Menu Off'	'Down' 'Enter'	'Left' 'Menu On'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command PictureMode ²	Value 'Presentation' 'Vivid' 'Infographic'	Value 'sRGB' 'User 1' 'User 2'	Value 'Bright' '3D'
# PictureMode example InterfaceName.Set('PictureMode', 'Presentation')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume	Value 0 to 10 in steps of 1		
# Volume example InterfaceName.Set('Volume', 10)			

¹This command is only available when PC signal (analog RGB) is selected.²This command will not function when 3D function is activated.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command 3DMode	Value 'Auto' 'Frame Packing' 'Inverter Disabled'	Value 'Top and Bottom' 'Side by Side' 'Inverter Enabled'	Value 'Frame Sequential' 'Sync Off'
# 3DMode examples InterfaceName.Update('3DMode') Value = InterfaceName.ReadStatus('3DMode') InterfaceName.SubscribeStatus('3DMode', None, FeedbackHandler)			
Command AspectRatio	Value '4:3' 'Auto'	Value '16:9' 'Real'	Value '16:10'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	

# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'PC 1 / YPbPr 1' 'HDMI 2 / MHL'	Value 'PC 2 / YPbPr 2' 'Video'	Value 'HDMI 1' 'S-Video'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Normal' 'Custom'	Value 'Economic'	Value 'Dimming'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command PictureMode	Value 'Presentation' 'Vivid' 'Infographic'	Value 'sRGB' 'User 1' 'User 2'	Value 'Bright' '3D'
# PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			

Command	Value
Volume	0 to 10 in steps of 1
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>	

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

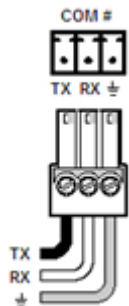
Data Bits: 8

Parity: None

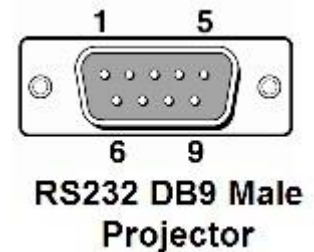
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface

Port Type:	Ethernet
Default Port:	8000
Logon Credentials Supported:	No
Multi-Connection	Undetermined
Capabilities:	
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Ensure that the following setting is set in the projector's OSD menu for the driver to work properly.

- System → Standby Settings → Enable Network Standby Mode: On

Appendix A. Set Commands

3D Mode Auto	\x0D*3d=auto#\x0D
3D Mode Frame Packing	\x0D*3d=fp#\x0D
3D Mode Frame Sequential	\x0D*3d=fs#\x0D
3D Mode Inverter Disabled	\x0D*3d=da#\x0D
3D Mode Inverter Enabled	\x0D*3d=iv#\x0D
3D Mode Side by Side	\x0D*3d=sbs#\x0D
3D Mode Sync Off	\x0D*3d=off#\x0D
3D Mode Top and Bottom	\x0D*3d=tb#\x0D
Aspect Ratio 16:10	\x0D*asp=16:10#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Real	\x0D*asp=REAL#\x0D
Audio Mute Off	\x0D*mute=off#\x0D
Audio Mute On	\x0D*mute=on#\x0D
Auto Image None	\x0D*auto#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input HDMI 1	\x0D*sour=hdm1#\x0D
Input HDMI 2 / MHL	\x0D*sour=hdm12#\x0D
Input PC 1 / YPbPr 1	\x0D*sour=RGB#\x0D
Input PC 2 / YPbPr 2	\x0D*sour=RGB2#\x0D
Input S-Video	\x0D*sour=svid#\x0D
Input Video	\x0D*sour=vid#\x0D
Lamp Mode Custom	\x0D*lampm=custom#\x0D
Lamp Mode Dimming	\x0D*lampm=dimming#\x0D
Lamp Mode Economic	\x0D*lampm=eco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Menu Off	\x0D*menu=off#\x0D
Menu Navigation Menu On	\x0D*menu=on#\x0D
Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D
Picture Mode 3D	\x0D*appmod=threed#\x0D

Global Scripter Module Communication Sheet

Picture Mode Bright	\x0D*appmod=bright#\x0D
Picture Mode Infographic	\x0D*appmod=infographic#\x0D
Picture Mode Presentation	\x0D*appmod=preset#\x0D
Picture Mode sRGB	\x0D*appmod=srgb#\x0D
Picture Mode User 1	\x0D*appmod=user1#\x0D
Picture Mode User 2	\x0D*appmod=user2#\x0D
Picture Mode Vivid	\x0D*appmod=vivid#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D
Volume 0	\x0D*vol=0#\x0D
Volume 10	\x0D*vol=10#\x0D

Appendix B. Update Commands

3D Mode	\x0D*3d=?#\x0D
Aspect Ratio	\x0D*asp=?#\x0D
Audio Mute	\x0D*mute=?#\x0D
Freeze	\x0D*freeze=?#\x0D
Input	\x0D*sour=?#\x0D
Lamp Mode	\x0D*lampm=?#\x0D
Lamp Usage	\x0D*ltim=?#\x0D
Picture Mode	\x0D*appmod=?#\x0D
Power	\x0D*pow=?#\x0D
Video Mute	\x0D*blank=?#\x0D
Volume	\x0D*vol=?#\x0D